ILLUSTRATIVE ENGAGEMENT-REVIEW PACK

Bounded Investigator / Skeptic Assessment

FireEvent FE-20190901-1ecb99d2e4 | Three structured rounds using the supplied audit evidence only

OBSERVED WINDOW	THERMAL OBSERVATIONS	PEAT CONTEXT
01 Sep - 03 Sep 2019	**6 linked FIRMS points**	**Mapped peat: yes**
04:55 UTC to 05:58 UTC	Mean FRP 10.34 MW; max 16.84 MW	100% footprint samples; 0.0 km nearest peat

REVIEW POSTURE

This is an evidence-grounded interpretation for human investigation. FIRMS observations do not independently establish a fire mechanism, ignition cause, actor, intent, wrongdoing, or responsibility. Scores are directional support labels, not probabilities or findings.

WORKING HYPOTHESES

H1 independent local ignition; H2 neighbouring surface propagation; H3 peat-mediated persistence or propagation; H4 multiple related land-management ignitions; H5 regional independent events under shared conditions; H6 other mechanism.

Observed and deterministic record

The record supports that a linked thermal-detection cluster was registered. It does not yet contain completed Fire Complexity or Investigation Priority enrichment.

EVIDENCE	WHAT IT STATES	USE / BOUNDARY
Chronology	2019-09-01T04:55Z to 2019-09-03T05:58Z.	Establishes the detection window; not continuous combustion.
Observations	6 linked FIRMS observations.	Thermal observations, not a direct count of separate fires.
FRP	Mean 10.34 MW; maximum 16.84 MW.	Thermal signal description; does not identify mechanism.
Stage-1 confidence	6/6 parseable; normalized mean 0.58.	Usable detection input, not causal attribution.
Nearby detections	14 outside this cluster within 10 km and 24 h.	Relationship worth testing; does not establish propagation.
Scope relation	EXTERNAL_CONTEXT agair private audit scope.	Contextual event only; not scope-linked attribution.
Routing state	Stage-1 AMBIGUOUS; evidence sufficiency PARTIAL.	Review-routing signal, explicitly not a finding.

Evidence IDs: OBSERVED_FE-20190901-1ecb99d2e4_chronology / _observations / _frp; DERIVED_STAGE1_FE-20190901-1ecb99d2e4_confidence / _max_frp / _observation_count / _nearby_detection_count; DERIVED_SCOPE_FE-20190901-1ecb99d2e4_relation; DERIVED_PRIORITY_FE-20190901-1ecb99d2e4_event_validity / _evidence_sufficiency.

ENVIRONMENTAL CONTEXT

Weather, peat and imagery availability

Environmental evidence supplies context. It does not identify ignition, prove a spread pathway, or establish peat combustion.

EVENT-WINDOW WEATHER

Mean 2m temperature 26.23; relative humidity 77.11%; precipitation 1.2 mm; 2 rainfall-free days. Mean 10m wind 11.05 at 151.8 deg; 100m wind 17.37 at 150.6 deg. Soil moisture: 0-7 cm 0.30, 7-28 cm 0.28, 28-100 cm 0.33. Total cloud cover 65.18%.

SEVEN-DAY ANTECEDENT

Mean 2m temperature 26.26; relative humidity 74.8%; precipitation 4.9 mm; 6 rainfall-free days. Mean 10m wind 11.31 at 150.2 deg; 100m wind 18.72 at 147.6 deg. Soil moisture: 0-7 cm 0.32, 7-28 cm 0.28, 28-100 cm 0.34. Total cloud cover 65.82%.

MAPPED PEATLAND

Footprint intersects mapped peat. 100.0% of footprint sample points are mapped peat; 80.3% of the 5.0 km buffer is mapped peat; nearest mapped peat is 0.0 km. Mapping alone does not show peat combustion or propagation.

AVAILABLE BEFORE / AFTER SCENES

Sentinel-2: pre 31 Aug 2019 02:16 UTC; post 05 Sep 2019 02:16 UTC. Sentinel-1: pre 30 Aug 2019 21:51 UTC; post 04 Sep 2019 21:59 UTC. No visual or backscatter interpretation was supplied, so this is availability evidence only.

Source groups: ENV_WEATHER_FE-20190901-1ecb99d2e4_current_* / _T-7d_* (Open-Meteo/ERA5); ENV_PEAT_FE-20190901-1ecb99d2e4_* (GPM 2022); ENV_IMAGERY_FE-20190901-1ecb99d2e4_Sentinel-1_* / Sentinel-2_* (Copernicus Data Space Ecosystem).

ROUND 1

Investigator - initial bounded reading

Generate interpretation from evidence IDs, not raw point dumps.

INVESTIGATOR

The event is a plausible multi-day thermal-detection cluster in a peat-dominated setting, with temporally and spatially nearby detections. This supports targeted investigation into persistence and local event relationships, but does not settle mechanism.

HYPOTHESIS	DIRECTIONAL SUPPORT	WHY
H1 independent local ignition	35 / PARTIAL	Local occurrence is compatible with local ignition, but no ignition-specific evidence exists.
H2 neighbouring surface propagation	40 / PARTIAL	14 nearby detections make a relationship worth testing; no tracked source, direction, or spread envelope.
H3 peat-mediated persistence / propagation	55 / PARTIAL	Mapped peat overlap and multi-day detection window support peat as relevant context, not proof of spread.
H4 related land-management ignitions	25 / INSUFFICIENT	No land-management, land-change, or coordinated-ignition evidence is present.
H5 regional independent events	45 / PARTIAL	Nearby detections plus similar weather conditions allow shared conditions as an alternative.
H6 other mechanism	50 / PARTIAL	AMBIGUOUS triage and uncompleted enrichment preserve a broad residual explanation.

Primary support: OBSERVED..._chronology / _observations / _frp; DERIVED_STAGE1..._nearby_detection_count; ENV_PEAT...; DERIVED_PRIORITY..._event_validity / _evidence_sufficiency.

ROUND 1

Skeptic - initial challenge

Separate evidence sufficiency from hypothesis support.

SKEPTIC

The record establishes detected heat observations and mapped peat overlap, but does not establish a single continuous fire, a propagation pathway, below-ground burning, or an ignition mechanism. The closest honest conclusion remains unresolved.

HYPOTHESIS	DIRECTIONAL SUPPORT	CHALLENGE
H1 independent local ignition	30 / PARTIAL	Possible, but no source, field observation, or ignition timing distinguishes it from alternatives.
H2 neighbouring surface propagation	20 / INSUFFICIENT	Counts within 10 km / 24 h do not show directional continuity or causal linkage.
H3 peat-mediated persistence / propagation	35 / PARTIAL	Peat mapping is spatial context; no depth, hydrology, smoke, or persistence measure was supplied.
H4 related land-management ignitions	10 / INSUFFICIENT	No activity, relationship, intent, or actor should be inferred.
H5 regional independent events	30 / PARTIAL	Weather is area-level context and does not demonstrate regional independence.
H6 other mechanism	60 / PARTIAL	The unanswered mechanism is strongest given partial sufficiency.

Critical gaps: all DERIVED_COMPLEXITY... components are NOT EVALUATED; key DERIVED_PRIORITY... components are NOT EVALUATED; imagery scenes have metadata only.

ROUND 2

Focused challenge and response

What survives adversarial review.

INVESTIGATOR REVISION

Reduce H2 because proximity does not establish spread. Retain H3 as a priority context because the footprint and nearby buffer are extensively mapped peat, while explicitly avoiding a peat-combustion finding.

EVIDENCE THAT STILL MATTERS

Six observations span more than two calendar days. Fourteen nearby detections justify relationship screening. Antecedent rain was limited to 4.9 mm over seven days with six rainfall-free days. Pre/post Sentinel-1 and Sentinel-2 scenes exist for targeted comparison.

SKEPTIC COUNTERWEIGHT

None of the remaining evidence converts proximity or dry-period context into propagation. The source-recorded soil-moisture values cannot be interpreted as a fire threshold without the data model, comparator, and validation.

DECISION CONSTRAINTS

Chronology is a detection span, not observed continuous burn duration. Weather is aggregate reanalysis context. Peat polygons do not establish depth, water-table state, or combustion. Scene IDs are not scene interpretation.

ROUND-2 CONVERGENCE

Both roles agree: the event merits evidence-led follow-up, peat deserves explicit environmental treatment, and no supplied input supports a mechanism, causal pathway, actor, or responsibility conclusion.

ROUND 3

Final structured position

Disagreement remains visible; no hypothesis is confirmed.

HYPOTHESIS	INVESTIGATOR	SKEPTIC	JOINT STATUS
H1 independent local ignition	30 / PARTIAL - compatible, unverified.	25 / PARTIAL - no discriminating evidence.	UNRESOLVED
H2 neighbouring surface propagation	30 / INSUFFICIENT - proximity warrants checking.	20 / INSUFFICIENT - no propagation evidence.	UNRESOLVED
H3 peat-mediated persistence / propagation	50 / PARTIAL - strong spatial peat context.	35 / PARTIAL - context is not combustion proof.	UNRESOLVED
H4 related land-management ignitions	10 / INSUFFICIENT	5 / INSUFFICIENT	NOT SUPPORTED
H5 regional independent events	40 / PARTIAL - competing explanation.	30 / PARTIAL - needs regional comparison.	UNRESOLVED
H6 other mechanism	55 / PARTIAL - residual category remains.	65 / PARTIAL - strongest fit for partial evidence.	UNRESOLVED

FINAL ROUTING VIEW

Evidence sufficiency remains PARTIAL. The audit package should show this review as AI interpretation alongside, and never instead of, the deterministic record and human review. No hypothesis is confirmed.

HUMAN REVIEW HANDOFF

Unresolved questions and targeted verification

The next steps are limited to evidence capable of differentiating mechanisms.

1. DOES THE IMAGE RECORD SHOW SPATIAL CONTINUITY OR MATERIAL CHANGE?

Retrieve and compare the listed pre/post Sentinel-1 and Sentinel-2 scenes over the event and nearby-detection area. Document method, areas reviewed and limitations; metadata alone is not interpretation. Evidence: ENV_IMAGERY..._Sentinel-2_pre_event / _post_event; ENV_IMAGERY..._Sentinel-1_pre_event / _post_event.

2. ARE NEARBY DETECTIONS CONTINUING LOCAL ACTIVITY, RELATED SPREAD, OR INDEPENDENT EVENTS?

Use time-ordered, spatially explicit analysis or higher-temporal-resolution observations to test continuity and direction. Compare with the event window and 10 km / 24 h relationship criterion. Evidence: OBSERVED..._chronology; DERIVED_STAGE1..._nearby_detection_count.

3. IS THERE DIRECT EVIDENCE OF PEAT-MEDIATED PERSISTENCE AT THIS LOCATION?

Seek validated peat depth, hydrology or water-table information, field observations, or other direct persistence indicators. Mapping and weather remain context until corroborated. Evidence: ENV_PEAT..._direct_intersection / _footprint_fraction / _buffer_fraction / _distance_to_peat; ENV_WEATHER..._current_soil_moisture_*.

KNOWN LIMITATIONS RETAINED

FIRMS observations are not a mechanism determination. Scope relation is EXTERNAL_CONTEXT. Fire Complexity and Investigation Priority components are not evaluated. No imagery-derived change metric, land-management activity, actor, or responsibility evidence was supplied.